

① a) 0 1 1 0 0 1 1 1 2

$$\begin{aligned}
 & \rightarrow 1 \cdot 2^0 = 1 \\
 & \rightarrow 1 \cdot 2^1 = 2 \\
 & \rightarrow 1 \cdot 2^2 = 4 \\
 & \rightarrow 0 \cdot 2^3 = 0 \\
 & \rightarrow 0 \cdot 2^4 = 0 \\
 & \rightarrow 1 \cdot 2^5 = 32 \\
 & \rightarrow 1 \cdot 2^6 = 64 \\
 & \rightarrow 0 \cdot 2^7 = 0
 \end{aligned}$$

103_{10}

② 00011001011010000111 2

1 9 6 8 7 16

③ 82_{10}

1000 64 32 16 8 4 2 1

01010010_2

④ $0x a160$

$$\begin{aligned}
 & 10 \cdot 16^3 \\
 & 15 \cdot 16^2 \\
 & 6 \cdot 16^1 \\
 & 0 \cdot 16^0
 \end{aligned}$$

$40960 + 3840 + 96 + 0 = 44896_{10}$

⑤ $0x a b c d$

$$\begin{aligned}
 & 1010 \quad 1011 \quad 1100 \quad 1101 \\
 & 1010 \quad 1011 \quad 1100 \quad 1101
 \end{aligned}$$

$1010 \ 1011 \ 1100 \ 1101_2$

⑥ 912342_{10}

$912342 / 16 = 57021$ resto 6 - 6

$57021 / 16 = 3563$ resto 13 - D

$3563 / 16 = 222$ resto 11 - B

$222 / 16 = 13$ resto 14 - E

$13 / 16 = 0$ resto 13 - D

$0x DEB66_{16}$

$$\textcircled{9} 131_{10} \quad \textcircled{128} \cancel{64} \cancel{32} \cancel{16} \cancel{8} \cancel{4} \textcircled{2} \textcircled{1}$$

$$10000011_2$$

2

$$\textcircled{a} \frac{1073741824}{1073741824} = 1 \text{ GiB}$$

$$\textcircled{b} \frac{1000 \quad 000 \quad 000}{1 \quad 000 \quad 000 \quad 000} = 190$$

$$\textcircled{c} 130 \cdot 1000 = 130 \text{ 000 bytes}$$

$$\textcircled{d} \frac{10 \quad 000 \quad 000 \quad 000}{1 \quad 073 \quad 741 \quad 824} = 11,18 \text{ GiB}$$

$$\textcircled{e} \frac{130 \quad 000 \quad 000}{1024^2} = 125,89 \text{ MiB}$$

$$\textcircled{f} \frac{11512}{8} = \frac{1439}{1000} = 1,439 \text{ GB}$$

$$\textcircled{3} 2^{30} - 1 = 4 \text{ 294 967 295 bytes}$$

$$\textcircled{4} 1 \text{ byte} = 8 \text{ bits} \quad \frac{40}{8} = 5 \text{ bytes}$$

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Operações Binárias

$$\begin{array}{r} \textcircled{a} \quad 11011101 \\ + 11010010 \\ \hline 110101112 \end{array}$$

$$\begin{array}{r} \textcircled{b} \quad 10110101 \\ - 00100110 \\ \hline 10011112 \end{array}$$

$$\begin{array}{r} \textcircled{c} \quad 101111 \\ \times \quad 11 \\ \hline 101111 \\ + 101111 \\ \hline 100011012 \end{array}$$

Represente em complemento a dois

$$\begin{array}{l} \text{msb} \\ \textcircled{a} \quad 110110002 \\ \text{inverte} - 00100111 \\ \text{soma } 1 - 001010002 = 40_{10} \end{array}$$

$$R: 11011000_2 = -40_{10}$$

$$\begin{array}{l} \text{msb} \\ \textcircled{b} \quad 000111012 \\ \hookrightarrow 000111012 \end{array}$$

$$\begin{array}{l} \text{msb} \\ \textcircled{c} \quad 11011102 \\ \text{inverte} - 00100001 \\ \text{soma } 1 - 001000102 = 34_{10} \end{array}$$

$$R: 11011101_2 = -34_{10}$$

Complemento a 1

$$\begin{array}{r} \textcircled{a} \quad 4_{10} + 18_{10} \quad 00000100 \quad 22_{10} \\ + 00010010 \\ \hline 000101102 \end{array}$$

$$\begin{array}{r} \textcircled{b} \quad 18_{10} - 14_{10} \\ 18 - 01110110 \\ 14 - 00001110 \\ C: 11110001 \\ \text{soma} \quad 01110110 \\ + 11110001 \\ \hline 10110011 \\ \hline 001101002 \end{array}$$

c) $-58_{10} + 49_{10}$

$58: 00111010$

$C: 11000101$

$49: 00110001$

normal 11000101

$+ 00110001$

11110110

inverse $00001001_2 = -9_{10}$

d) $102_{10} - 12_{10}$

$102: 01100110$

$12: 00001100$

$C: 11110011$

normal 01100110

$+ 11110011$

101011001

$\rightarrow 1$

01011010_2

90_{10}